

Compliance and policy landscape for a UCL SEIOT thesis conducted remotely from India

A UCL MSc student planning **90 days of remote thesis work from India** faces a web of interacting compliance constraints across immigration, academic policy, data protection, and institutional rules. The most critical finding is that **Graduate Route visa eligibility is at serious risk** under Immigration Rules Appendix Graduate GR 6.1, which requires the full duration of a 12-month course to have been studied in the UK. (www) UCL's Study Away policy technically permits up to three months for PGT students, but this may not "shield" the absence under UKVI rules, since remote thesis writing from home does not qualify as a "permitted study abroad programme" (which requires a partner institution overseas). The student faces a narrow compliance corridor where academic approval, visa rules, and data governance must all align — and several of these constraints pull in opposite directions.

1. UCL permits up to three months of Study Away for PGT students

UCL Academic Manual Chapter 3, Section 3.5.2 (2025–26) states: *"You may also be permitted to study away from UCL on academic grounds which are not part of the standard delivery of a programme or module e.g., collecting data or conducting research. You must seek permission to do this from your Departmental Tutor/Programme Organiser, agreeing the period of study away, and the location. You cannot do this for more than three months."* Section 3.5.3 adds that PGT students working on their dissertation during the summer vacation *"may be based at UCL or work remotely — individual arrangements should be agreed with your supervisor."* (ucl)

The approval chain runs: (1) apply to the Departmental Tutor or Programme Leader, providing location and reason; (2) the department signs off on the form; (3) the signed form is sent to UCL's **Student Immigration Compliance Team**. The form is called the **"Study Away from UCL form"** (downloadable as `study_away_from_ucl_2022.doc`) from UCL's student status pages). (ucl) No specific processing time is published, and no additional documentation beyond the form itself is described on the public-facing page.

The three categories of absence are formally distinct. **Authorised Absence** covers short-term illness or extenuating circumstances (self-certified up to 7 days; **maximum 60 calendar days** for Student visa holders (UCL) during term time). (ucl) (UCL) **Study Away** covers academic work conducted away from campus — the student remains enrolled and actively studying.

Interruption of Study pauses registration entirely, typically for one year minimum, (UCL) and triggers withdrawal of Student visa sponsorship. (ucl) (UCL) A **one-week personal trip** falls under Authorised Absence (self-certified). A **three-month remote thesis period** falls under Study Away — but **90 days sits exactly at the stated maximum**, and whether "up to three months" includes or excludes the boundary is ambiguous.

No formal host institution is required for Study Away that is not part of standard programme delivery. Section 3.5.2 only requires agreement on the period and location. Independent remote research is implicitly acceptable. However, the actual form (a .doc file) may contain fields not described on the webpage.

Remote presentations for PGT MSc theses are not governed by a published UCL-wide policy. PGT programmes at UCL generally do not include a formal viva — assessment is primarily through the written dissertation. For PGR degrees, UCL explicitly permits remote vivas ("*may take place at UCL, or remotely*"). (UCL) (UCL) The SEIOT programme includes a **Designing Sensor Systems Showcase** at UCL East, which appears to be a module-specific presentation event rather than a thesis defense. (UCL) The specific SEIOT thesis assessment format — including whether it has an oral component and whether remote participation is permitted — could not be determined from public sources and must be confirmed with the programme leader.

Unknowns / verify with: UCL Student Immigration Compliance Team (immigration.compliance@ucl.ac.uk) for visa implications of Study Away; SEIOT programme leader for thesis presentation format; Student Records for processing time of Study Away applications.

Sources: UCL Academic Manual Chapter 3, Section 3 (2025–26):

<https://www.ucl.ac.uk/academic-manual/chapters/chapter-3-registration-framework-taught-programmes/section-3-ucl-student-attendance-policy>; Study Away page:

<https://www.ucl.ac.uk/students/student-status/changes-your-studies/study-away-ucl>; Doctoral School viva guidance: <https://www.ucl.ac.uk/doctors-school/regulations/examiners/examiner-viva-guidance>

2. Graduate Route eligibility is the highest-risk compliance issue

There is no explicit maximum number of days a Student visa holder may spend outside the UK in the Immigration Rules themselves. The compliance framework operates through **sponsor engagement monitoring** rather than a hard absence cap. UCL, as a Higher Education Provider with a track record of compliance, monitors **academic engagement** (not physical attendance at "contact points," which is the standard for non-HEPs). The widely cited **60-day threshold** appears in institutional policies (ucl) derived from UKVI Sponsor Guidance, which requires sponsors to report when a student defers or is unable to engage for more than 60 days.

(Uklegalguides)

The critical constraint is **Immigration Rules Appendix Graduate, GR 6.1**, which governs Graduate Route eligibility. For courses lasting **12 months or less**, the applicant must have held Student permission "*for the full duration of course*" during which "*all study took place in the UK (apart from permitted study abroad programmes)*." For courses **longer than 12 months**, only 12 months of in-UK study is required. (www) The gov.uk public guidance clarifies: "*Studying your*

course in the UK means that you were in the UK when your education provider needed you to be there — for example, to go to lectures or meet with a tutor." (GOV.UK)

This is the pivotal issue. A standard UCL MSc runs September to September (approximately 12 months). If classified as 12 months or less, the student must have studied for the full duration in the UK. A **90-day absence during the dissertation period could disqualify Graduate Route eligibility** unless it constitutes a "permitted study abroad programme." The Graduate Route Caseworker Guidance (v5.0, November 2025) defines such programmes narrowly: study must be *"at a partner institution outside the UK"* that is *"integral and assessed"* as part of the course. (www) **Remote thesis writing from home in India — with no partner institution — does not meet this definition.**

The Graduate Route also requires the applicant to **be physically in the UK** on the date of application (GR 1.2(d)) (www) and to have had their sponsor notify UKVI of successful course completion (GR 4.2–4.3). (GOV.UK) There is **no requirement for continuous presence in the final stretch** specifically, (GOV.UK) but the "study in the UK" requirement under GR 6.1 applies across the entire relevant period.

UCL's reporting obligations include notifying UKVI when a student interrupts studies, withdraws, or is withdrawn for non-engagement. UCL's Student visa responsibilities page states that *"authorised absences during term time must not exceed 60 calendar days"* (UCL) and that exceeding this likely requires an interruption of study, *"which would result in the withdrawal of your Student visa."* (UCL) Whether UCL-approved Study Away (where the student remains engaged and studying) is treated differently from Authorised Absence for sponsor reporting purposes is not definitively addressed in public guidance — this is a key ambiguity.

Border re-entry during Study Away should not itself create a compliance issue if the Student visa remains valid and the student is still sponsored. However, Border Force may question the nature and duration of absence.

Unknowns / verify with: UCL Immigration Compliance Team (critical — ask whether Study Away shields the absence for GR 6.1 purposes); UKVI (whether remote dissertation work counts as "studying in the UK" when no face-to-face attendance is required); the CAS to confirm exact course duration classification.

Sources: Immigration Rules Appendix Graduate (updated 5 March 2026):

<https://www.gov.uk/guidance/immigration-rules/immigration-rules-appendix-graduate>;

Graduate Route Caseworker Guidance v5.0 (November 2025):

<https://assets.publishing.service.gov.uk/media/68f62f24f5d433238a14c7df/Graduate+route.pdf>;

Sponsor Guidance Document 2 (v04/2026):

<https://www.gov.uk/government/publications/student-sponsor-guidance/sponsorship-duties-accessible>; UCL Student visa responsibilities: <https://www.ucl.ac.uk/study/immigration-and-visas/student-visas/student-visa-responsibilities>

<https://www.ucl.ac.uk/study/immigration-and-visas/student-visas/student-visa-responsibilities>

3. Ethics review likely not required for aggregate data, but DPIA and IP agreements are

UCL requires ethics approval for all research involving human participants and/or their data.

The central guidance (updated December 2025) states: *"At UCL, all research (including pilot studies) involving human participants and/or their data or tissue requires ethical approval."* (UCL)

However, UCL explicitly exempts **"minimal risk"** research from ethics review, (UCL) including *"use of anonymised records and datasets in the public domain"* and *"use of information available in the public domain."* (UCL)

Aggregated historical traffic volume data (vehicle counts per hour per intersection) that contains no personal identifiers would likely fall under the minimal risk exemption and **not require full ethics committee review** — provided no individual vehicles or persons can be re-identified from the data. The key test is whether the specific dataset could identify living individuals (e.g., if it inadvertently contains license plate data or GPS traces). If the data is genuinely aggregate, ethics review is not required, though UCL recommends consulting the Research Ethics Service if uncertain. (UCL)

Consulting municipal officials about traffic patterns crosses a different threshold. UCL's guidance defines human participant research as *"any research involving direct interaction with other people, or that uses data that have been collected from individuals."* (UCL) If the student conducts structured interviews or consultations with municipal officials generating new primary data, this **triggers ethics review** — though a **low-risk route** would apply if no vulnerable groups, sensitive topics, or deception are involved. (UCL) Simply corresponding with authorities to obtain pre-existing datasets (administrative data acquisition) would not.

A Data Protection Impact Assessment (DPIA) may be triggered even if no personal data is processed. UCL's DPIA template lists triggers including *"when applying innovative use or applying technological solutions, e.g. 'internet of things'"* and *"when data is transferred outside the European Union."* (UCL) Given this project involves IoT/edge computing and data from India, the student should run UCL's **DPIA screener** (<https://ucl.logicgate.com/>) to determine whether a full DPIA is required. If the traffic data is truly anonymous, a DPIA is not required on the basis that *"personal data is not being processed"* (UCL) — but the innovative technology trigger may still apply.

For industry-collaborated theses, UCL's EEE-IXN (Industry Exchange Network) policy states that projects *"should not require a Non-Disclosure Agreement (NDA)"* and that NDAs are considered *"only in exceptional circumstances for MSc projects"* — any NDA must go through **UCL's legal vetting** and must not interfere with assessment. (UCL) On IP, UCL's policy recognizes the student as owner of IP produced during registration, **unless** the work is part of a larger collaboration or funded by an external organisation, in which case UCL may negotiate IP assignment. (UCL) The EEE-IXN policy for industry projects states that *"IPR for outputs of work*

undertaken during a project will be assigned to the company." (UCL) A **collaboration agreement** is needed for projects involving external research collaborators, (UCL) and a **Material Transfer Agreement** is required when "materials or data are transferred between UCL and another organisation." (UCL) UCL provides template agreements for sharing anonymised data.

Unknowns / verify with: UCL Research Ethics Service (weekly drop-ins Tuesdays 2:30–3:30pm); (UCL) UCL Data Protection Office (data-protection@ucl.ac.uk) for DPIA screener results; UCL Research Innovation Services (Contracts) for collaboration agreement with Microsoft; SEIOT programme supervisor for IP arrangement specifics.

Sources: UCL ethics guidance (December 2025): <https://www.ucl.ac.uk/research-innovation-services/compliance-and-assurance/research-ethics-service/do-i-need-ethical-approval>; UCL DPIA guidance: <https://www.ucl.ac.uk/data-protection/guidance-staff-students-and-researchers/practical-data-protection-guidance-notice/data-protection>; EEE-IXN policy: <https://ucl.ac.uk/electronic-electrical-engineering/collaborate/industry-supervised-msc-undergraduate-projects>; UCL IP policy (September 2019): <https://www.ucl.ac.uk/enterprise/students/understand-your-intellectual-property-ip-rights>

4. Aggregate traffic counts fall outside UK GDPR, but data flow architecture matters

UK GDPR applies to all processing of personal data by UCL regardless of data subject nationality, because UCL is a UK establishment. The ICO confirms: "Everyone in the UK benefits from the protection of the UK GDPR. It is the physical location of the individual that matters, not where they are from." (Information Commissioner'...) However, the **critical threshold** is whether the data constitutes "personal data" at all.

License plates constitute personal data (an identifier linkable to a registered vehicle owner). **CCTV facial images** constitute personal data and potentially **special category data** (biometric) if processed for identification. **Aggregated vehicle counts per hour** that cannot identify individual vehicles or persons are **not personal data** and fall entirely outside UK GDPR scope. (ICO) UK GDPR Recital 26 confirms: "The principles of data protection should therefore not apply to anonymous information, namely information which does not relate to an identified or identifiable natural person." (ICO) The ICO applies a "**motivated intruder**" test — whether a determined person could re-identify individuals. (Pinsent Masons) (ICO) Hourly aggregate counts at a busy intersection pass this test easily.

For the three-way data flow (India → UCL → Microsoft): importing aggregate anonymous data from India to the UK is **not a "restricted transfer"** under UK GDPR (restricted transfers only cover outbound UK-to-non-UK flows of personal data). (ICO) (ICO) If any personal data were involved and transferred from UCL to Microsoft in the US, this **would** be a restricted transfer requiring a lawful transfer mechanism. Microsoft may be covered by the **UK Extension to the**

EU-US Data Privacy Framework (partial adequacy, effective October 2023) if Microsoft is DPF-certified — this should be verified. (Information Commissioner'...) Alternative mechanisms include the **International Data Transfer Agreement (IDTA)**, in force since 21 March 2022. (ICO)

India does not have UK GDPR adequacy status. (ucl) India is listed as a "longer-term priority" for UK adequacy assessment. (GRCI Law Blog) The **Digital Personal Data Protection Act 2023 (DPDPA)** was enacted on 11 August 2023, (Meity) with DPDP Rules 2025 finalized on 13 November 2025 under phased implementation. **Phase 3** — covering consent requirements, data principal rights, and **cross-border transfers under Section 16** — does not become enforceable until **13 May 2027**. (Meity) The DPDPA adopts a blacklist model: transfers are permitted unless the government restricts them to specific countries. **No countries have been blacklisted** as of April 2026. The DPDPA defines personal data as "*any data about an individual who is identifiable by or in relation to such data*" — aggregate traffic counts that cannot identify individuals fall outside its scope.

Synthetic data generated from real-world traffic parameters occupies legally uncertain territory. (ICO) The ICO recognizes synthetic data as a privacy-enhancing technology (Syntheticus) but notes that **the process of generating synthetic data from personal data is itself processing** requiring a lawful basis. (Timelex) If the synthetic data is generated from statistical distributions (not individual-level records) and cannot be linked to specific individuals, it is more likely to qualify as non-personal data — but this requires a **case-by-case risk assessment**. (Ga4gh) The DPDPA does not explicitly address synthetic data.

Unknowns / verify with: ICO helpline for synthetic data classification in this specific use case; UCL Data Protection Office regarding DPIA for the three-way data flow; US Department of Commerce DPF website (dataprivacyframework.gov) to confirm Microsoft's current DPF certification status.

Sources: ICO territorial scope guidance: <https://ico.org.uk/for-organisations/uk-gdpr-guidance-and-resources/personal-information-what-is-it/who-does-the-uk-gdpr-apply-to/>; ICO adequacy regulations (updated 15 January 2026): <https://ico.org.uk/for-organisations/uk-gdpr-guidance-and-resources/international-transfers/adequacy-regulations/>; ICO anonymisation guidance: <https://ico.org.uk/for-organisations/uk-gdpr-guidance-and-resources/data-sharing/anonymisation/>; DPDPA text: <https://www.meity.gov.in/static/uploads/2024/06/2bf1f0e9f04e6fb4f8fef35e82c42aa5.pdf>; DPDP Rules phased implementation: MeitY notification G.S.R. 843(E), 13 November 2025

5. Microsoft has no MSc-specific thesis program, but open resources are substantial

No publicly documented Microsoft program exists specifically for MSc-level industry thesis collaborations. The major academic programs — Microsoft Research Fellowship (\$27,000 for

European fellows; 2026 deadline passed 2 December 2025), the Dissertation Grant (up to \$25,000, US/Canadian PhD students only), (Microsoft) and the PhD Fellowship (nomination-based, 3rd-year PhD) — all target doctoral or postdoctoral researchers. The **Microsoft Research Cambridge lab internship** is the closest fit, as it *"mostly offer[s] internships to Masters or PhD students"* (Microsoft) for 12-week paid research positions where work can be incorporated into a thesis. (Alex Lu)

Azure for Students provides **\$100 in Azure credits** (no credit card required, (Microsoft Azure) renewable annually) covering most Azure services including AI/ML tools. Microsoft's description explicitly mentions *"powerful small language models (SLMs) with groundbreaking performance at low cost and low latency."* (Microsoft Azure) (Microsoft News) The **Phi model family** (Phi-1 through Phi-4-mini-reasoning) is fully **open source under MIT License**, available on Hugging Face, GitHub, and Ollama. Microsoft explicitly positions Phi for edge deployment: *"Think of car computers, PCs without Wi-Fi, traffic systems, smart sensors on a factory floor."* (Microsoft News)

On IP, Microsoft's policy for academic institutions provides *"appropriate, royalty-free covenants under its patents"* for non-commercial research. (Microsoft) Fellowship recipients are *"not subject to IP restrictions."* (Microsoft) For data sharing, Microsoft has published template agreements through its Open Data Campaign, including the **Data Use Agreement for Data Commons (DUA-DC)** — a template for multi-party data sharing through API-enabled databases. The DUA-DC assumes data does not include personal/private/proprietary data by default but includes provisions for modification where PII is involved. (Microsoft)

For PII protection in edge computing, Microsoft provides **Azure IoT Edge security** (hardware root of trust, secure enclaves, encryption at rest/in transit), (Microsoft Learn) **Confidential Computing** (Trusted Execution Environments for data-in-use protection), (Microsoft Learn) and open-source tools including **Microsoft Presidio** (de-identification for text and images) and **SmartNoise** (differential privacy). No public documentation specifically addresses PII handling in academic edge computing pilots.

Unknowns / verify with: Microsoft Research Cambridge (for internship timelines and whether thesis collaboration outside a formal internship is possible); the specific Microsoft collaborator/supervisor on the project for IP arrangement expectations; dataprivacyframework.gov for Microsoft's DPF certification status.

Sources: Microsoft Research Academic Programs: <https://www.microsoft.com/en-us/research/academic-programs/>; Azure for Students: <https://azure.microsoft.com/en-us/free/students>; Phi models: <https://azure.microsoft.com/en-us/products/phi>; Microsoft IP licensing: <https://www.microsoft.com/en-us/legal/intellectualproperty/tech-licensing/policy>; DUA-DC template: <https://cdn-dynmedia-1.microsoft.com/is/content/microsoftcorp/microsoft/msc/documents/presentations/CSR/DUA-DC-0-1.pdf>; Microsoft Responsible AI: <https://www.microsoft.com/en-us/ai/principles-and-approach>

6. India offers multiple data access routes, but none are fast

The most promising formal channels for accessing Indian traffic data are:

- **India Urban Data Exchange (IUDX)** at <https://iudx.org.in/> — operational in **55+ Smart Cities**, developed by IISc Bangalore with the Smart Cities Mission. (IUDX) Provides authenticated API-based access using NGSI-LD standards. IUDX explicitly lists academia as a stakeholder: *"Institutions abroad can also access unique India datasets and solve unique problems for Indian cities."* (IUDX) Traffic-relevant datasets exist (e.g., Surat transit management). (IUDX) A sandbox environment is available for testing. (IUDX)
- **Smart Cities Open Data Portal** at <https://smartcities.data.gov.in/> — hosts **3,566+ catalogs** from 100 Smart Cities (Smart Cities) under the **Government Open Data License - India (GODL-India, February 2017)**, which grants a worldwide, royalty-free license for commercial and non-commercial use. (Data) Traffic datasets exist for cities including Coimbatore.
- **National Open Government Data Platform** at <https://data.gov.in/> — free access, no registration required.
- **City Data Officers (CDOs)** — all 100 Smart Cities have appointed CDOs (lee) whose contact emails are publicly listed at <https://dsc.smartcities.gov.in/cdo>. CDOs are the designated single point of contact for external data requests.

The Right to Information Act 2005 is a statutory right but carries a **critical limitation: only Indian citizens may file RTI applications** (Section 3). A UCL student who is not an Indian citizen would need an Indian collaborator to file on their behalf. RTI applications cost ₹10 (~£0.10), with a **30-day statutory response deadline**. However, RTI is designed for targeted information requests, not bulk data extraction — and a Public Information Officer may refuse to provide information if it would *"disproportionately divert the resources of the public authority"* (Section 7(9)), though the Delhi High Court has ruled that bulkiness alone is not valid grounds for denial. (Onlinerti)

Cities with accessible traffic-adjacent data include Bengaluru (traffic signal timing data via OpenCity.in; the **UVH-26 dataset** of 26,678 annotated CCTV images from Bengaluru Traffic Police released by IISc), Surat (transit management data on IUDX), (IUDX) and Coimbatore (traffic data on Smart Cities portal). A significant gap exists: **Indian open traffic datasets are primarily image/video-based** (object detection), not time-series flow/speed data comparable to METR-LA or PEMS.

No standardized timeline exists for obtaining formal research cooperation agreements (MoUs) with Indian municipal bodies. The DataSmart Cities Strategy envisions (Ministry of Urban Develop...) **City Data Alliances** including academic institutions, but most cities are still developing their City Data Policies. Informal channels through CDOs or Smart City SPV CEOs are likely faster than formal MoU routes, but no documented timelines were found.

Unknowns / verify with: IUDX team (contact@iudx.org.in) for API access process and available traffic datasets; specific city CDOs for data availability; Indian academic collaborator for RTI filing if needed.

Sources: IUDX: <https://iudx.org.in/>; Smart Cities Open Data Portal: <https://smartcities.data.gov.in/>; CDO directory: <https://dsc.smartcities.gov.in/cdo>; GODL-India: <https://smartcities.data.gov.in/government-open-data-license-india>; DataSmart Cities Strategy: https://smartcities.gov.in/DataSmart_Cities_Strategy; RTI Act: <https://rti.gov.in/>; RTI fee rules: <https://legalaffairs.gov.in/rti/fee-required-under-rti-act>

7. UCL accommodation requires the student to find their own replacement

UCL-managed accommodation operates under a **Licence Agreement** (not a tenancy), giving the student a contractual right to occupy but no legal interest in the property. (UCL) The 2025–26 contract for PG students runs from **20 September 2025 to 12 September 2026**. (UCL) Early termination requires submitting a **Notice to Quit (NTQ)** through the UCL Accommodation Portal, (UCL) but the critical condition is: *"It is your responsibility to find a suitable replacement student to take over your Accommodation."* (UCL) (UCL) If a replacement is found, liability ends when that student accepts the booking. **If no replacement is found, the student remains liable for fees until the contract end date.** (UCL) (UCL) The replacement must be a UCL student not currently in UCL-allocated accommodation, (UCL) at the same level of study (PG for PG). (UCL) **Subletting is strictly prohibited** — the regulations state *"you must not allow any other person to occupy or share your Accommodation"* (UCL) and breach can result in licence termination. A **£400 prepayment** made at offer acceptance may be forfeited upon early termination. (UCL) (UCL) Credit balances are refunded within **28 days** from the contract end date to a UK bank account. (UCL)

For a **23 May departure** with a PG contract ending 12 September, the student would need a replacement willing to take over for approximately **112 days** — more viable than a 28-day UG remainder but still challenging in late May when many students are completing their own programmes.

Private tenancies in Camden are governed by the specific agreement terms. Under a standard **Assured Shorthold Tenancy (AST)**, the tenant is responsible for rent for the entire fixed term. Early exit requires either a **break clause** (GOV.UK) (not standard for tenancies under two years), **mutual surrender** (landlord agreement), (Sandwell Council) or finding a replacement tenant

acceptable to the landlord. (www) The **Tenant Fees Act 2019** permits early termination fees (GOV.UK) but only reflecting the landlord's actual loss. (Camden Council) Subletting an AST typically requires express landlord consent; most student tenancies prohibit it. (Tenant-rights) Under a **licence agreement** (common for student HMOs), the specific terms govern — the licensee generally need only give "reasonable notice," typically the length of one rental period. (www) Deposits are capped at **5 weeks' rent** (for annual rent under £50,000), (GOV.UK) must be protected in an approved scheme within 30 days, (Camden Council) and must be returned within **10 days** of both parties agreeing the amount. (GOV.UK)

Platforms for room handovers include the **UCL Students' Union Flatmate Finder** (<https://studentsunionucl.org/flatmate-finder>), (Students' Union UCL) **SpareRoom** (645 rooms currently listed near UCL), (Spareroom) **The Student Room** forums (active UCL accommodation swap threads), and various UCL Facebook groups (unofficial). Realistic turnaround for a private room in Camden is **1-4 weeks** on SpareRoom; for UCL halls replacements in late May, prospects are significantly more challenging and UCL "*cannot guarantee that a suitable replacement student will be found.*" (ucl)

Sources: UCL 2025-26 Licence Agreement: <https://www.ucl.ac.uk/accommodation/fees-guidance-and-payment/general-regulations/ucl-accommodation-2025-2026> (updated 17 September 2025); UCL moving out: <https://www.ucl.ac.uk/accommodation/current-residents/moving-out>; GOV.UK tenancy guidance: <https://www.gov.uk/private-renting-tenancy-agreements/how-to-end-your-tenancy>; Deposit protection: <https://www.gov.uk/tenancy-deposit-protection>; Tenant Fees Act Schedule 1: <https://www.legislation.gov.uk/ukpga/2019/4/schedule/1>

8. SUMO is the validated platform for synthetic traffic data, and the thesis topic is novel

SUMO (Simulation of Urban Mobility) is the clear first-choice platform, (eclipse) currently at version 1.26.0 under Eclipse Public License v2.0. It provides a full **Python API via TraCI** (Traffic Control Interface), (PubMed Central) imports real-world road networks from **OpenStreetMap**, and supports microscopic multi-modal simulation with unlimited network size. A **2024 Lund University MSc thesis** directly combined SUMO simulation with edge computing for real-time traffic signal optimization using Differential Evolution — validating this exact approach at thesis level. (LUP Student Papers) Multiple peer-reviewed publications (2023-2026) use SUMO-generated synthetic data for edge computing traffic research, establishing clear academic precedent.

CityFlow is the strongest alternative, purpose-built for **reinforcement learning** traffic signal control. Licensed under Apache 2.0, it provides seamless C++/Python integration via pybind11 and runs **20× faster** than SUMO on large-scale networks. An extended version, **CityFlowER** (ECML-PKDD 2024), adds embedded ML models. **Aimsun** is commercial (ResearchGate) (€20,000-25,000/year) and not recommended unless an academic licence is secured. **Flow** (UC

Berkeley, MIT License) wraps SUMO with RL capabilities via RLlib/OpenAI Gym (Berkeley) (Flow-project) but shows limited recent development activity. **MATSim** (GPLv2, Java-based) excels at city-scale demand modeling but is less suited for direct edge-device SLM testing. (Wikipedia) (CIVITAS)

The most-cited public traffic datasets are:

- **METR-LA**: 207 highway sensors, Los Angeles, March–June 2012, 5-minute speed readings — the benchmark for spatiotemporal traffic prediction
- **PEMS-BAY**: 325 sensors, San Francisco Bay Area, January–May 2017, 5-minute intervals — co-benchmark with METR-LA
- **UTD19: 23,541 detectors across 40+ cities worldwide** (including London), ~5 billion vehicle detections, published in Scientific Reports 2019 — the largest multi-city urban dataset and most relevant for diverse urban contexts (Ethz +2)
- **NYC TLC Trip Data**: billions of taxi/FHV trip records (2009–present), open data (GitHub)
- **Uber Movement: discontinued**, but archive data remains on Kaggle for some cities including Bangalore (QGIS Tutorials and Tips)

Indian traffic datasets are primarily **image/video-based** (IDD from IIIT Hyderabad: 10,000 annotated images; (liit) UVH-26 from IISc/Bengaluru Traffic Police: 26,678 annotated CCTV images; DATS_2022: 10,000+ images from Maharashtra). **No publicly available Indian urban traffic flow time-series data** comparable to METR-LA/PEMS was found — validating the need for synthetic data generation via SUMO if aggregate flow data cannot be obtained from municipal authorities.

No existing publication from UCL, Imperial College, or Microsoft combines edge SLMs specifically with traffic signal optimization. Microsoft explicitly lists "*traffic systems*" as a target for Phi deployment, (Microsoft News) and **Phi-4-mini** is described as "*ideal for lightweight deployment on edge or mobile systems*" for IoT applications "*in environments with limited computing power and network access.*" (Microsoft Azure) (Microsoft Community Hub) UCL has published on edge computing for IoT (Federated Active Transfer Learning), edge-cloud collaboration with small edge models, and decentralised mobility-aware resource allocation — but not in the specific SLM-for-traffic intersection. **This confirms the thesis topic's novelty.**

Sources: SUMO: <https://eclipse.dev/sumo/>; CityFlow: <https://github.com/cityflow-project/CityFlow>; UTD19: <https://utd19.ethz.ch/>; Lund University thesis: <https://lup.lub.lu.se/student-papers/record/9188899/file/9188900.pdf>; Microsoft Phi: <https://azure.microsoft.com/en-us/products/phi>; UCL edge-cloud paper: <https://discovery.ucl.ac.uk/id/eprint/10194347/>; UVH-26: <https://iisc.ac.in/events/aim-iisc-announces-public-release-of-uvh-26-dataset-and-vision-models-for-indian-urban-traffic/>; Phi-3

Conclusion

The compliance landscape reveals three dominant constraints that shape feasibility. First, **Graduate Route visa eligibility under GR 6.1 is the single highest-risk factor** — a 90-day absence during a 12-month MSc almost certainly falls outside the narrow "permitted study abroad programme" exception, and whether remote dissertation work counts as "studying in the UK" when no face-to-face attendance is required remains genuinely unresolved in published UKVI guidance. This must be verified directly with UCL's Immigration Compliance Team before any travel commitment. Second, **data protection compliance is manageable** if the student works exclusively with genuinely aggregated traffic data — aggregate vehicle counts fall outside both UK GDPR and DPDPA scope, eliminating most cross-border transfer complications. Third, **the technical fallback is robust**: SUMO-based synthetic traffic data combined with Microsoft's open-source Phi-4-mini on edge devices is a validated, precedented approach that does not depend on obtaining municipal data at all.

The absence of any published work combining edge SLMs with traffic signal optimization — at UCL, Imperial, or Microsoft Research — confirms this thesis would occupy a genuinely novel position. The open-source availability of both SUMO and the Phi model family under permissive licenses means the technical infrastructure carries no licensing barriers. The primary remaining uncertainties are immigration-related (GR 6.1 interpretation), institutional (SEIOT-specific thesis assessment format), and procedural (Indian municipal data access timelines). Each of these should be resolved through direct consultation with the flagged offices before the planned 23 May departure.